

# The Defect Vector Is a Directional Derivative

## Cross-Prompt Verification of the Transport Identity, a Cheap Row-Transport Corollary, and a Corrected Magnitude–Instability Correlation

Abdallah Khemais  
*ISITCOM, University of Sousse*

### Abstract

A companion paper [1] shows that the per-branch coefficient  $q_j$  of a backward readout depends on the read site only through one vector per pair of sites: the whole site-to-site difference is a single linear functional  $\langle \beta_j^{(t)}, u_{s,t} \rangle$ , and the entire row  $\{q_j^{(s)}\}_j$  is generated by one vector  $h_s$ . That paper already proves, and we do not re-derive, that  $h_s$  is itself a directional derivative of the network’s own forward map:  $h_s = J_{\Phi_{s \rightarrow t}}(y_s) g^{(s)}$ , hence estimable from an independent forward-only computation by a centered finite difference with a proven  $O(\varepsilon^2)$  Lagrange-remainder error, and that this was checked on the production model on one 16-token prompt to a median error of  $1.10 \times 10^{-5}$  at  $\varepsilon = 10^{-2}$ . This note adds four things, one of them a correction. First, a cost corollary the companion paper does not state: given the ablation sweep already run at the deep site, the shallow site’s entire row on the shared branches is recoverable from two forward passes, not  $B(t)$  further backward passes. Second, we extend the numerical check from one prompt to seven, spanning 5 to 16 tokens, at the same site pair; the aggregate median error at  $\varepsilon = 10^{-2}$  is  $1.01 \times 10^{-5}$ , in agreement with the single-prompt figure to three significant figures. (The six new prompts also supplied the cross-prompt round-off measurement now reported directly in [1]: the implied constant  $C_2 := \varepsilon \times \text{error}$  varies by a factor of 4.25 across independent prompts, not the 15% its single-prompt, multi-step-size check alone would suggest, though the seven-prompt median stays within 8% of the single-precision floor.) Third, we recall briefly, without re-deriving, the companion paper’s own site-dependence measurement at scale (22 prompts, 33,880 site pairs, median symmetric deviation 0.027, ninetieth percentile 0.262, 6.4% sign flips). Fourth, and this is the correction: a conversational estimate from a narrow earlier look had put the correlation between a branch’s causal magnitude and its cross-site ranking instability at Spearman  $\rho \approx 0.80$ – $0.91$ . Tested properly, on 154 (prompt, site-pair) combinations across eight site-depth pairs and 22 prompts, the raw correlation has median  $\rho = 0.43$  and the scale-normalized correlation, which removes a mechanical artifact of large-magnitude branches having more absolute room to deviate, has median  $\rho = -0.51$ : the earlier estimate is not supported by either measurement, and the sign itself is not stable across the normalization. We report this as the real result, not the guess. We also record, honestly and without speculation, that a root-site dominance anomaly analogous to one seen on a toy model was tested twice on the production model and produced no usable data either time; it remains open.

## 1 Setting and recap

We use the branch-gate construction, notation and results of [1] without restating its proofs: a branch gate  $\varepsilon_j$  on a join  $y = x + f(x)$  leaves the forward pass exactly unchanged while scaling the backward vector–Jacobian product across that join, the readout  $g^{(s)}$  at a site  $s$  is the true

backward value with all gates open, and for each bypassable branch  $j$  the split  $g^{(s)} = \beta_j^{(s)} + \sigma_j^{(s)}$  into the part routed through  $j$  and the part routed around it gives the exact coefficient  $q_j^{(s)} := \langle \beta_j^{(s)}, g^{(s)} \rangle / \|g^{(s)}\|^2$ , computable from  $B+1$  backward passes with no finite-difference step. This construction sits in the same lineage as activation patching and the circuit-discovery methods built on it, which isolate a component's causal contribution by selective ablation and compare the perturbed and unperturbed readouts [2]; here the ablation is over residual-stream branches specifically, with an exact closed form for the coefficient rather than an empirical search over components.

The result this note builds on is the transport identity of [1]: for two read sites  $s, t$  on one stream with  $t$  nearer the objective,  $\text{cone}(t) \subseteq \text{cone}(s)$ , and  $T$  the ordered product of the gate factors strictly between them, so that  $g^{(s)} = Tg^{(t)}$  with all gates open, the site-to-site difference of the shared coefficients collapses to one linear functional of a single vector,

$$q_j^{(s)} - q_j^{(t)} = \langle \beta_j^{(t)}, u_{s,t} \rangle \quad (j \in \text{cone}(t)),$$

with  $u_{s,t}$  independent of  $j$ , and the entire row  $\{q_j^{(s)}\}_j$  is in fact generated by the single vector  $h_s := T^\top g^{(s)}$  through

$$q_j^{(s)} = \frac{\langle \beta_j^{(t)}, h_s \rangle}{\langle g^{(t)}, h_s \rangle} \quad (j \in \text{cone}(t)). \quad (1)$$

Everything below concerns  $h_s$ : what it is, how it is estimated independently, what that estimate is worth in cost, and how it was checked.

## 2 The defect vector as a directional derivative

The companion paper already identifies  $h_s$  with a directional derivative of the network's own forward computation and proves the finite-difference error bound this note verifies numerically; we restate it, adapted to fix notation for Section 3 below, and cite [1] for the proof rather than reproduce it in full.

**Corollary 1** (The defect vector as a directional derivative, [1]). *Let  $y_s$  denote the residual-stream activation at the join immediately downstream of the branch defining site  $s$ ,  $y_t$  the same for  $t$ , and  $\Phi_{s \rightarrow t}$  the network's own forward map from  $y_s$  to  $y_t$  — the ordinary composition of the joins strictly between them, independent of every gate parameter since the forward pass is untouched by construction. Then*

$$T = J_{\Phi_{s \rightarrow t}}(y_s)^\top, \quad \text{so} \quad h_s = T^\top g^{(s)} = J_{\Phi_{s \rightarrow t}}(y_s) g^{(s)}, \quad (2)$$

*the directional derivative of  $\Phi_{s \rightarrow t}$  at  $y_s$  in the direction  $g^{(s)}$ . Consequently, for  $u := g^{(s)} / \|g^{(s)}\|$  and  $\varepsilon > 0$  small enough that  $\Phi_{s \rightarrow t}$  is three-times continuously differentiable on  $[y_s - \varepsilon u, y_s + \varepsilon u]$ ,*

$$\left\| \|g^{(s)}\| \cdot \frac{\Phi_{s \rightarrow t}(y_s + \varepsilon u) - \Phi_{s \rightarrow t}(y_s - \varepsilon u)}{2\varepsilon} - h_s \right\| \leq \frac{\|g^{(s)}\| \varepsilon^2}{6} \sup_{\xi \in [y_s - \varepsilon u, y_s + \varepsilon u]} \|D^3 \Phi_{s \rightarrow t}(\xi)\|. \quad (3)$$

*Sketch, adapted from [1].* Each bypassable join contributes  $I + \varepsilon_j K_j$  with  $K_j = J_{f_j}^\top$ , so at  $\varepsilon_j = 1$  it is  $(I + J_{f_j})^\top$ , the transpose of the join's own forward Jacobian; a mandatory join contributes

$C_j = J_{f_j}^\top$ , likewise a transposed forward Jacobian.  $T$  is the ordered product of exactly these factors, hence the transpose, in reverse order, of the ordered product of the corresponding untransposed local Jacobians — which is  $J_{\Phi_{s \rightarrow t}}(y_s)$  by the chain rule,  $\Phi_{s \rightarrow t}$  being the ordinary forward composition of those same joins. This gives (2); (3) is the standard third-order remainder bound for a centered difference, the zeroth- and second-order terms cancelling and the first-order term reducing to  $2\varepsilon J_{\Phi_{s \rightarrow t}}(y_s)u$ .  $\square$

Write  $\hat{h}_s$  for the finite-difference estimate inside the norm of (3) and, substituting it into (1),

$$\hat{q}_j^{(s)} := \frac{\langle \beta_j^{(t)}, \hat{h}_s \rangle}{\langle g^{(t)}, \hat{h}_s \rangle}, \quad j \in \text{cone}(t), \quad (4)$$

for the finite-difference-predicted coefficient. Section 3 reports how well  $\hat{q}_j^{(s)}$  agrees with the independently, exactly measured  $q_j^{(s)}$ .

**Corollary 2** (Cheap row transport). *Suppose the exact ablation sweep of [1] has already been run at the deep site  $t$ , giving  $\beta_j^{(t)}$  and  $g^{(t)}$  for every  $j \in \text{cone}(t)$  at a cost of  $B(t) + 1$  backward passes, and that  $g^{(s)}$  is available from the ordinary backward pass at  $s$ . Then the entire row  $\{q_j^{(s)}\}_{j \in \text{cone}(t)}$  —  $B(t)$  further exact coefficients — is recoverable to the accuracy of (3) from two additional forward evaluations of  $\Phi_{s \rightarrow t}$ , via (4), with no further backward pass at  $s$  at all.*

*Proof.* Equation (1) expresses every  $q_j^{(s)}$ ,  $j \in \text{cone}(t)$ , as a fixed function of the already-available  $\beta_j^{(t)}$ ,  $g^{(t)}$  and the single vector  $h_s$ ; nothing in (1) depends on  $j$  except through  $\beta_j^{(t)}$ , which is already in hand from the  $t$ -side sweep. Corollary 1 supplies  $\hat{h}_s$ , an estimate of  $h_s$ , from  $\Phi_{s \rightarrow t}(y_s \pm \varepsilon u)$  alone — two forward passes, computed once and reused across every  $j$  — with no backward pass through the joins between  $s$  and  $t$ . Substituting  $\hat{h}_s$  for  $h_s$  in (1) gives (4), whose error is controlled termwise by (3) composed with the (fixed,  $j$ -independent) linear map  $h \mapsto \langle \beta_j^{(t)}, h \rangle / \langle g^{(t)}, h \rangle$ .  $\square$

The exact alternative — closing each of the  $B(t)$  gates at  $s$  in turn and re-running the backward pass at  $s$  — costs  $B(t)$  backward passes through the whole graph between the objective and  $s$ ; the row-transport estimate costs two forward passes between  $y_s$  and  $y_t$  only, a fraction of the graph, and does not scale with  $B(t)$  at all. The trade-off recalls the one that motivates cheap approximations to activation patching elsewhere in the causal-tracing literature: attribution patching [3] and AtP\* [4] likewise replace an exhaustive per-component ablation sweep with a single cheap gradient- or activation-based linear approximation, accepting a controlled approximation error for a large constant-factor speedup. Corollary 2 is the same trade in a forward-only, JVP-based form specific to the row-transport identity of [1]. Whether that estimate is accurate enough to replace the exact sweep is exactly the empirical question of Section 3, and the answer there is conditional: it depends on  $\varepsilon$ , and on the round-off floor identified in Section 3, not on some hypothetical exactness of (4).

### 3 Cross-prompt verification

The single-prompt check reported in [1] used  $s$  at the layer-1 attention site ( $\text{cone } B = 56$ ) and  $t$  at the layer-13 attention site ( $|\text{cone}(t)| = 32$ ) on the trained 1.5B-parameter model, one prompt of 16 tokens, and found a median absolute error of  $1.10 \times 10^{-5}$  at  $\varepsilon = 10^{-2}$  between  $\hat{q}_j^{(s)}$  and the directly,

| Prompt (first words)                   | Tokens | err. median at $\varepsilon = 10^{-2}$ | $\ g^{(s)}\ $          | Source           |
|----------------------------------------|--------|----------------------------------------|------------------------|------------------|
| "If a train travels 60 miles. ..."     | 16     | $1.0962 \times 10^{-5}$                | $3.513 \times 10^{-1}$ | .log, 2026-08-15 |
| "The capital of France is"             | 5      | $2.2473 \times 10^{-5}$                | $1.149 \times 10^0$    | _results.txt     |
| "SELECT name FROM users. ..."          | 7      | $5.2830 \times 10^{-6}$                | $1.190 \times 10^0$    | _results.txt     |
| "Question: Who wrote Hamlet?..."       | 9      | $6.3459 \times 10^{-6}$                | $1.507 \times 10^0$    | _results.txt     |
| "The capital of the country famous..." | 11     | $9.0711 \times 10^{-6}$                | $6.365 \times 10^{-1}$ | _results.txt     |
| "The capital of the country where..."  | 13     | $1.1829 \times 10^{-5}$                | $4.626 \times 10^{-1}$ | _results.txt     |
| "It was a bright cold day..."          | 14     | $1.1147 \times 10^{-5}$                | $8.977 \times 10^{-1}$ | _results.txt     |

Table 1: The seven distinct prompts underlying the aggregate figure below, all at  $s = \text{layer-1}$  attention output,  $t = \text{layer-13}$  attention output ( $|\text{cone}(t)| = 32$ ), on the same trained 1.5B-parameter checkpoint. Six prompts and the aggregate summary are from `notebook/bench_eps_jvp_transport_check_results.txt`; the 16-token prompt is from `notebook/bench_eps_jvp_transport_check.log` and from [1] itself.

independently measured  $q_j^{(s)}$ , over the 32 shared branches. We re-ran the identical check, same site pair, same corrective measures — the perturbed node’s forward rule must be suspended to force recomputation, and  $y_s$  must be the residual-stream join output rather than the raw branch output — on six new prompts the earlier check did not use, spanning 5 to 14 tokens (Table 1), all recorded in `notebook/bench_eps_jvp_transport_check_results.txt` (2026-08-18). Combined with the 16-token prompt of [1], preserved in `notebook/bench_eps_jvp_transport_check.log` (2026-08-15), this is seven distinct prompts at one site-depth pair.

Pooling branch-level errors across the six new prompts, the aggregate median error at  $\varepsilon = 10^{-2}$  is  $1.0109 \times 10^{-5}$ , against a median  $|q_j^{(s)}|$  around  $8 \times 10^{-2}$  on each prompt individually — three significant figures of agreement with the disjoint single-prompt figure of [1], obtained days earlier from an unrelated prompt. At  $\varepsilon = 10^{-3}$  and  $\varepsilon = 10^{-4}$  the aggregate median error rises to  $4.5762 \times 10^{-5}$  and  $6.2153 \times 10^{-4}$ , an empirical slope of about  $-0.9$  rather than the  $+2$  that (3)’s truncation term alone would give: as in [1], this is round-off dominating truncation over the tested range, not a failure of the bound. One coefficient in every prompt’s 32 keeps a fixed error of essentially 1 at every  $\varepsilon$ ; this is not noise but the exact algebraic identity  $\hat{q}_t^{(s)} = 1$  that  $j = t$  forces in (4) whenever  $\sigma_t^{(t)} = 0$ , which [1] derives from the mandatory-branch corollary and which recurs unchanged here.

The round-off constant  $C_2 := \varepsilon \times \text{error}$  this implies is a property of the seven prompts pooled, not of any one of them: fixing  $\varepsilon = 10^{-2}$  and reading  $C_2$  off each prompt of Table 1 independently gives values ranging over a factor of 4.25, median  $1.10 \times 10^{-7}$ , still within 8% of the single-precision floor  $1.19 \times 10^{-7}$ . This is the cross-prompt measurement now reported directly in [1], which also corrects that paper’s earlier, narrower claim (consistency checked only across step sizes on one prompt); we do not reproduce the table here and refer to it there.

## 4 Magnitude does not predict cross-site instability

For scale, and measured, not re-measured here: [1] reports, across all 22 prompts of its own multi-prompt sweep, 1,540 pairs of adjacent sites sharing a branch per prompt, 33,880 pairs in all, a symmetric deviation  $|q^{(i)} - q^{(i+1)}|/(|q^{(i)}| + |q^{(i+1)}|)$  with median 0.027 and ninetieth percentile 0.262, and 6.4% of pairs of opposite sign; we do not reproduce that sweep here, only reuse its site set and prompts below. A question raised in earlier, narrower session work was whether a branch’s own causal magnitude at a reference site predicts how unstable its coefficient

| $t$ (site position) | branches in cone( $t$ ) | $\rho_{\text{raw}}$ (mean $\pm$ std) | $\rho_{\text{norm}}$ (mean $\pm$ std) |
|---------------------|-------------------------|--------------------------------------|---------------------------------------|
| 8                   | 49                      | $0.510 \pm 0.176$                    | $-0.455 \pm 0.126$                    |
| 13                  | 44                      | $0.426 \pm 0.141$                    | $-0.524 \pm 0.139$                    |
| 16                  | 41                      | $0.484 \pm 0.169$                    | $-0.460 \pm 0.142$                    |
| 21                  | 36                      | $0.413 \pm 0.153$                    | $-0.514 \pm 0.164$                    |
| 25                  | 32                      | $0.362 \pm 0.152$                    | $-0.521 \pm 0.155$                    |
| 33                  | 24                      | $0.395 \pm 0.184$                    | $-0.487 \pm 0.193$                    |
| 45                  | 12                      | $0.414 \pm 0.224$                    | $-0.368 \pm 0.345$                    |

Table 2: Per-site-pair mean  $\pm$  std over prompts, for the two Spearman correlations. The  $t = 55$  row is omitted: too few prompts leave  $\geq 4$  shared branches at that depth to report a mean.  $\rho_{\text{raw}}$  stays positive and moderate across every depth tested;  $\rho_{\text{norm}}$  stays negative and moderate across every depth tested. Neither approaches the conversational estimate at any depth.

is across sites — whether large- $|q_j|$  branches are also the ones whose rank shifts most when the read site changes. A first look, on a handful of site pairs, suggested a strong relationship, estimated conversationally at Spearman  $\rho \approx 0.80$ – $0.91$ . That estimate was not a measurement, and it does not survive one; more generally, magnitude is not a reliable stand-in for the stability of an attribution, a concern raised for saliency and feature-importance methods elsewhere in the interpretability literature [5].

We tested it properly using the reference site  $s = 1$  (layer-1 attention output, full cone) against eight deeper sites  $t \in \{8, 13, 16, 21, 25, 33, 45, 55\}$ , over all 22 prompts of the existing multi-prompt matrix. For each prompt and site pair, on the shared branches  $j \in \text{cone}(t)$ , define

$$\begin{aligned}
\text{magnitude}(j) &:= |q_j^{(s)}|, \\
\text{deviation}(j) &:= |q_j^{(s)} - q_j^{(t)}|, \\
\text{deviation}_{\text{norm}}(j) &:= \frac{|q_j^{(s)} - q_j^{(t)}|}{\frac{1}{2}(|q_j^{(s)}| + |q_j^{(t)}|) + 10^{-12}},
\end{aligned}$$

and compute the Spearman correlation of magnitude against each of the two deviation measures across the branches of that (prompt, site-pair) instance. Pairs with fewer than four shared branches are dropped (chiefly the deepest site,  $t = 55$ , whose cone is small); 154 of the  $8 \times 22 = 176$  possible (prompt, site-pair) combinations remain (recorded in [notebook/bench\\_eps\\_rank\\_correlation\\_magnitude\\_deviation\\_results.txt](#)).

Over all 154 combinations,  $\rho_{\text{raw}}$  has median 0.430 (range  $-0.126$  to  $0.858$ ) and  $\rho_{\text{norm}}$  has median  $-0.506$  (range  $-0.895$  to  $0.357$ ). Both are far from the guessed  $0.80$ – $0.91$ , and they disagree with each other in sign at the median. The reason is not a contradiction but a scale artifact, and it is worth stating precisely because it is the mechanism, not a caveat.  $\text{deviation}(j)$  is an absolute difference: a branch with large  $|q_j^{(s)}|$  has, mechanically, more room to move in absolute terms before the topology-driven bound of [1]’s two-scalar closed form caps it, so magnitude and raw deviation are positively coupled almost by construction, independent of any real instability.  $\text{deviation}_{\text{norm}}$  divides by the branch’s own scale and removes most of that coupling; what remains is negative, meaning that, if anything, the largest-magnitude branches are *more* stable in *relative* terms across sites, not less. Neither reading supports treating causal magnitude as a proxy for cross-site rank instability, and the earlier  $0.80$ – $0.91$  estimate should be read as an artifact of too narrow a look, not as a result this note confirms.

## 5 What this note does not settle

A root-site dominance anomaly, analogous to one observed on a toy model during earlier session work, was tested on the production model to see whether it reproduces at that scale. Two attempts were made (`notebook/bench_eps_qwen_root_site_dominance.jl` and its companion results file); both stalled before producing usable per-branch data. The recorded output confirms only the correction gates — bit-exact forward re-wiring at  $\varepsilon \equiv 1$ , a bit-exact gradient at the root site under that re-wiring, and a sanity check that ablating one branch does change the root-site gradient — and stops there. We report this as an open question, not as a finding: we do not know whether the toy-model anomaly reproduces on the production model, and nothing in what follows should be read as evidence either way.

## Discussion

The four additions of this note vary in weight, and it is worth being plain about which. Corollary 2 is a genuine, if narrow, cost observation: given the ablation sweep already run at a deep site, a shallow site’s row on the shared branches is two forward passes away rather than  $B(t)$  backward passes away, a saving that grows with cone size. The cross-prompt verification of Section 3 strengthens [1]’s single-prompt check from an existence proof to a small statistic; the round-off scope correction this check produced is reported directly in the updated [1] rather than as a finding of this note. Section 4 is the piece most worth flagging: a conversational estimate did not survive a properly powered test, the two natural normalizations of the same question disagree in sign, and the honest conclusion is that causal magnitude is not a reliable predictor of cross-site instability on this model, in either direction, once the scale artifact is controlled for. Section 5 is a record of two stalled attempts and nothing more; it should not be over-read as either confirming or ruling out the anomaly it was built to test.

## References

- [1] A. Khemais. *The Branch-Order Decomposition of a Backward Readout: An Exact Identity for the Mean Branch Order, Its Two-Scalar Form, and the Obstruction to a Closed Form*. Preprint, 2026.
- [2] A. Conmy, A. N. Mavor-Parker, A. Lynch, S. Heimersheim, A. Garriga-Alonso. *Towards Automated Circuit Discovery for Mechanistic Interpretability*. Advances in Neural Information Processing Systems 36, 2023.
- [3] A. Syed, C. Rager, A. Conmy. *Attribution Patching Outperforms Automated Circuit Discovery*. NeurIPS ATTRIB Workshop (arXiv:2310.10348), 2023.
- [4] J. Kramár, T. Lieberum, R. Shah, N. Nanda. *AtP\*: An Efficient and Scalable Method for Localizing LLM Behaviour to Components*. arXiv:2403.00745, 2024.
- [5] A. Ghorbani, A. Abid, J. Zou. *Interpretation of Neural Networks Is Fragile*. Proceedings of the AAAI Conference on Artificial Intelligence, 2019.